# Electrical Transient Analyzer Program

## Short-Circuit Analysis

### IEC 60909 Standard

### 3-Phase Fault Currents

| | Swing | V-Control | Load | Total |
|---|---|---|---|---|
| Number of Buses: | 1 | 0 | 22 | 23 |

| | XFMR2 | XFMR3 | Reactor | Line/Cable/ Busway | Impedance | Tie PD | Total |
|---|---|---|---|---|---|---|---|
| Number of Branches: | 7 | 0 | 0 | 15 | 0 | 0 | 22 |

| | Synchronous Generator | Power Grid | Synchronous Motor | Induction Machines | Lumped Load | Total |
|---|---|---|---|---|---|---|
| Number of Machines: | 0 | 1 | 0 | 0 | 6 | 7 |

| | |
|---|---|
| System Frequency: | 50.00 |
| Unit System: | Metric |
| Project Filename: | data_center_study_bootcamp |
| Output Filename: | D:\Software\ETAP\data_center_study_bootcamp\Untitled.SI1S |

## Adjustments

| Tolerance | Apply Adjustments | Individual /Global | Percent |
|---|---|---|---|
| Transformer Impedance: | Yes | Individual | |
| Reactor Impedance: | Yes | Individual | |
| Overload Heater Resistance: | No | | |
| Transmission Line Length: | No | | |
| Cable / Busway Length: | No | | |

| Temperature Correction | Apply Adjustments | Individual /Global | Degree C |
|---|---|---|---|
| Transmission Line Resistance: | Yes | Individual | |
| Cable / Busway Resistance: | Yes | Individual | |

## Bus Input Data

| Bus | | | | | Initial Voltage | |
|---|---|---|---|---|---|---|
| ID | Type | Nom. kV | Base kV | Sub-sys | %Mag. | Ang. |
| 20 kV MVP-L1-CSR-A | Load | 20.000 | 20.000 | 1 | 100.00 | 0.00 |
| Bus PLN-A | SWNG | 20.000 | 20.000 | 1 | 100.00 | 0.00 |
| Bus2 | Load | 0.400 | 0.400 | 1 | 100.00 | 0.00 |
| Bus3 | Load | 20.000 | 20.000 | 1 | 100.00 | 0.00 |
| Bus4 | Load | 0.400 | 0.400 | 1 | 100.00 | 0.00 |
| Bus5 | Load | 20.000 | 20.000 | 1 | 100.00 | 0.00 |
| Bus6 | Load | 0.400 | 0.400 | 1 | 100.00 | 0.00 |
| Bus7 | Load | 0.400 | 0.400 | 1 | 100.00 | 0.00 |
| Bus8 | Load | 0.400 | 0.410 | 1 | 100.00 | 0.00 |
| Bus9 | Load | 0.400 | 0.410 | 1 | 100.00 | 0.00 |
| Bus10 | Load | 0.400 | 0.410 | 1 | 100.00 | 0.00 |
| Bus11 | Load | 11.000 | 11.282 | 1 | 100.00 | 0.00 |
| Bus12 | Load | 0.400 | 0.410 | 1 | 100.00 | 0.00 |
| Bus13 | Load | 0.400 | 0.410 | 1 | 100.00 | 0.00 |
| Bus15 | Load | 20.000 | 20.000 | 1 | 100.00 | 0.00 |
| Bus16 | Load | 0.400 | 0.410 | 1 | 100.00 | 0.00 |
| Bus17 | Load | 11.000 | 11.282 | 1 | 100.00 | 0.00 |
| Bus19 | Load | 11.000 | 11.282 | 1 | 100.00 | 0.00 |
| Bus20 | Load | 11.000 | 11.282 | 1 | 100.00 | 0.00 |
| LDMP-L2-C1 | Load | 0.400 | 0.410 | 1 | 100.00 | 0.00 |
| MV-GSB-01 | Load | 11.000 | 11.282 | 1 | 100.00 | 0.00 |
| MVSB-L2-CH1 | Load | 11.000 | 11.282 | 1 | 100.00 | 0.00 |
| MVSB-L4-A1 | Load | 11.000 | 11.282 | 1 | 100.00 | 0.00 |

23 Buses Total

All voltages reported by ETAP are in % of bus Nominal kV.
Base kV values of buses are calculated and used internally by ETAP.

## Line/Cable/Busway Input Data

| Line/Cable/Busway | | | ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway) | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | Length | | | | | | |
| ID | Library | Size | Adj. (m) | % Tol. | #/Phase | T (°C) | R | X | Y |
| Cable1 | 20NCUS3 | 300 | 83.0 | 0 | 2 | 75 | 0.07621 | 0.09270 | 0.0000000 |
| Cable2 | 20NCUS3 | 120 | 20.0 | 0 | 1 | 75 | 0.18694 | 0.10600 | 0.0000000 |
| Cable3 | 1.0NCUS3 | 300 | 100.0 | 0 | 8 | 75 | 0.07621 | 0.06970 | 0.0000000 |
| Cable4 | 20NCUS3 | 120 | 20.0 | 0 | 1 | 75 | 0.18694 | 0.10600 | 0.0000000 |
| Cable5 | 1.0NCUS3 | 300 | 100.0 | 0 | 8 | 75 | 0.07621 | 0.06970 | 0.0000000 |
| Cable6 | 11MCUS3 | 240 | 72.7 | 0 | 1 | 75 | 0.09347 | 0.09000 | 0.0001495 |
| Cable7 | 1.0MCUN1 | 300 | 31.0 | 0 | 4 | 75 | 0.07420 | 0.11270 | 0.0000000 |
| Cable8 | 11MCUS3 | 240 | 72.2 | 0 | 1 | 75 | 0.09347 | 0.09000 | 0.0001495 |
| Cable9 | 1.0MCUN1 | 300 | 20.0 | 0 | 13 | 75 | 0.07420 | 0.11270 | 0.0000000 |
| Cable10 | 11MCUS3 | 240 | 68.5 | 0 | 1 | 75 | 0.09347 | 0.09000 | 0.0001495 |
| Cable11 | 20NCUS3 | 300 | 61.6 | 0 | 1 | 75 | 0.07621 | 0.09270 | 0.0000000 |
| Cable12 | 11MCUS3 | 300 | 35.3 | 0 | 1 | 75 | 0.07630 | 0.08740 | 0.0001646 |
| Cable13 | 0.6MCUN1 | 300 | 34.0 | 0 | 13 | 75 | 0.14764 | 0.14108 | 0.0000000 |
| Cable14 | 11MCUS1 | 120 | 24.6 | 0 | 1 | 75 | 0.18694 | 0.11800 | 0.0001109 |
| Cable15 | 20NCUS3 | 95 | 20.0 | 0 | 1 | 75 | 0.23558 | 0.10900 | 0.0000000 |

Line / Cable / Busway resistances are listed at the specified temperatures.

## 2-Winding Transformer Input Data

| Transformer | Rating | | | | | Z Variation | | | % Tap Setting | | Adjusted |
|---|---|---|---|---|---|---|---|---|---|---|---|
| ID | MVA | Prim. kV | Sec. kV | % Z | X/R | + 5% | - 5% | % Tol. | Prim. | Sec. | % Z |
| T-Loadbank01 | 5.500 | 11.000 | 0.400 | 7.15 | 8.50 | 0 | 0 | 0 | 0 | 0 | 7.1500 |
| TRF-HV-L1-A1 | 10.000 | 20.000 | 11.000 | 12.00 | 6.00 | 0 | 0 | 0 | -2.500 | 0 | 12.0000 |
| TRF-L2-CH1 | 4.200 | 11.000 | 0.400 | 7.50 | 8.50 | 0 | 0 | 0 | 0 | 0 | 7.5000 |
| TRF-L8-A1 | 4.000 | 11.000 | 0.400 | 7.50 | 8.50 | 0 | 0 | 0 | 0 | 0 | 7.5000 |
| TRF-L8-A3 | 1.400 | 11.000 | 0.400 | 7.50 | 6.00 | 0 | 0 | 0 | 0 | 0 | 7.5000 |
| TRF-L10-A1 | 2.800 | 20.000 | 0.400 | 7.50 | 6.00 | 0 | 0 | 0 | 0 | 0 | 7.5000 |
| TRF-L10-A3 | 2.000 | 20.000 | 0.400 | 7.50 | 6.00 | 0 | 0 | 0 | 0 | 0 | 7.5000 |

## Branch Connections

| CKT/Branch | | Connected Bus ID | | % Impedance, Pos. Seq., 100 MVAb | | | |
|---|---|---|---|---|---|---|---|
| ID | Type | From Bus | To Bus | R | X | Z | Y |
| T-Loadbank01 | 2W XFMR | MV-GSB-01 | Bus16 | 13.81 | 117.42 | 118.23 | |
| TRF-HV-L1-A1 | 2W XFMR | Bus15 | Bus17 | 18.30 | 109.79 | 111.30 | |
| TRF-L2-CH1 | 2W XFMR | Bus11 | Bus8 | 18.94 | 160.98 | 162.09 | |
| TRF-L8-A1 | 2W XFMR | Bus19 | Bus13 | 19.89 | 169.02 | 170.19 | |
| TRF-L8-A3 | 2W XFMR | Bus20 | Bus10 | 79.96 | 479.78 | 486.40 | |
| TRF-L10-A1 | 2W XFMR | Bus5 | Bus6 | 42.06 | 252.35 | 255.83 | |
| TRF-L10-A3 | 2W XFMR | Bus3 | Bus4 | 58.88 | 353.29 | 358.16 | |
| Cable1 | Cable | Bus PLN-A | 20 kV MVP-L1-CSR-A | 0.08 | 0.10 | 0.12 | |
| Cable2 | Cable | 20 kV MVP-L1-CSR-A | Bus3 | 0.09 | 0.05 | 0.11 | |
| Cable3 | Cable | Bus4 | Bus2 | 59.54 | 54.45 | 80.68 | |
| Cable4 | Cable | 20 kV MVP-L1-CSR-A | Bus5 | 0.09 | 0.05 | 0.11 | |
| Cable5 | Cable | Bus6 | Bus7 | 59.54 | 54.45 | 80.68 | |
| Cable6 | Cable | MVSB-L4-A1 | Bus20 | 0.53 | 0.51 | 0.74 | |
| Cable7 | Cable | Bus10 | Bus9 | 34.17 | 51.89 | 62.13 | |
| Cable8 | Cable | MVSB-L4-A1 | Bus19 | 0.53 | 0.51 | 0.74 | |
| Cable9 | Cable | Bus13 | Bus12 | 6.78 | 10.30 | 12.33 | |
| Cable10 | Cable | MVSB-L4-A1 | MVSB-L2-CH1 | 0.50 | 0.48 | 0.70 | |
| Cable11 | Cable | 20 kV MVP-L1-CSR-A | Bus15 | 0.12 | 0.14 | 0.18 | |
| Cable12 | Cable | Bus17 | MVSB-L4-A1 | 0.21 | 0.24 | 0.32 | |
| Cable13 | Cable | Bus8 | LDMP-L2-C1 | 22.94 | 21.92 | 31.73 | |
| Cable14 | Cable | MVSB-L2-CH1 | Bus11 | 0.36 | 0.23 | 0.43 | |
| Cable15 | Cable | MV-GSB-01 | MVSB-L4-A1 | 0.37 | 0.17 | 0.41 | |

## Power Grid Input Data

| Power Grid | Connected Bus | Rating | | % Impedance 100 MVA Base | | |
|---|---|---|---|---|---|---|
| ID | ID | MVASC | kV | R | X" | R/X |
| PLN A | Bus PLN-A | 150.000 | 20.000 | 6.63358 | 66.33581 | 0.10 |

Total Connected Power Grids ( = 1 ): 150.000 MVA

## Lumped Load Input Data

| Lumped Load | | | | | | | | Motor Loads | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Lumped Load | Connected Bus | Rating | | | % Load | | | Loading | | % Impedance Machine Base | | | |
| ID | ID | kVA | kV | Amp | % PF | MTR | STAT | kW | kvar | R | X" | R/X" | MW/PP |
| LDMP-L2-CH1 | LDMP-L2-C1 | 770.0 | 0.400 | 1111.40 | 95.00 | 80 | 20 | 585.2 | 192.3 | 6.46 | 15.37 | 0.42 | 0.59 |
| Loadbank G1 | Bus16 | 5000.0 | 0.400 | 7216.88 | 95.00 | 80 | 20 | 3800.0 | 1249.0 | 5.96 | 14.18 | 0.42 | 3.80 |
| UDB-L10-A1 | Bus7 | 1000.0 | 0.400 | 1443.38 | 95.00 | 80 | 20 | 760.0 | 249.8 | 5.96 | 14.18 | 0.42 | 0.76 |
| UDB-L10-A2 | Bus9 | 380.0 | 0.400 | 548.48 | 95.00 | 80 | 20 | 288.8 | 94.9 | 5.96 | 14.18 | 0.42 | 0.29 |
| UDB-L10-A3 | Bus2 | 1000.0 | 0.400 | 1443.38 | 95.00 | 80 | 20 | 760.0 | 249.8 | 5.96 | 14.18 | 0.42 | 0.76 |
| UDB-L10-A4 | Bus12 | 480.0 | 0.400 | 692.82 | 95.00 | 80 | 20 | 364.8 | 119.9 | 5.96 | 14.18 | 0.42 | 0.36 |

Total Connected Lumped Loads ( = 6 ): 8630.0 kVA

## SHORT-CIRCUIT REPORT

3-Phase fault at bus:

**20 kV MVP-L1-CSR-A**

| | | | |
|---|---|---|---|
| Nominal kV | = | 20.000 | |
| Voltage c Factor | = | 1.10 | (User-Defined) |
| Peak Value | = | 12.546 | kA Method C |
| Steady State | = | 4.324 | kA rms |

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| 20 kV MVP-L1-CSR-A | Total | 0.00 | 0.744 | -5.222 | 7.0 | 5.274 |
| Bus PLN-A | 20 kV MVP-L1-CSR-A | 0.19 | 0.434 | -4.302 | 9.9 | 4.324 |
| Bus3 | 20 kV MVP-L1-CSR-A | 0.01 | 0.050 | -0.126 | 2.5 | 0.135 |
| Bus5 | 20 kV MVP-L1-CSR-A | 0.01 | 0.053 | -0.131 | 2.5 | 0.141 |
| Bus15 | 20 kV MVP-L1-CSR-A | 0.04 | 0.206 | -0.663 | 3.2 | 0.694 |
| PLN A | Bus PLN-A | 110.00 | 0.434 | -4.302 | 9.9 | 4.324 |
| Bus4 | Bus3 | 16.80 | 0.050 | -0.126 | 2.5 | 0.135 |
| Bus6 | Bus5 | 12.54 | 0.053 | -0.131 | 2.5 | 0.141 |
| Bus17 | Bus15 | 27.48 | 0.206 | -0.663 | 3.2 | 0.694 |

### Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 5.231 | 7.283 | 5.067 |
| 0.02 | 5.208 | 6.363 | 3.656 |
| 0.03 | 5.166 | 5.765 | 2.559 |
| 0.04 | 5.124 | 5.428 | 1.792 |
| 0.05 | 5.071 | 5.246 | 1.343 |
| 0.06 | 5.042 | 5.131 | 0.953 |
| 0.07 | 5.013 | 5.059 | 0.677 |
| 0.08 | 4.985 | 5.008 | 0.480 |
| 0.09 | 4.958 | 4.970 | 0.341 |
| 0.10 | 4.931 | 4.938 | 0.258 |
| 0.15 | 4.872 | 4.872 | 0.048 |
| 0.20 | 4.815 | 4.815 | 0.009 |
| 0.25 | 4.761 | 4.761 | 0.002 |
| 0.30 | 4.761 | 4.761 | 0.000 |

3-Phase fault at bus: **Bus PLN-A**

Nominal kV = 20.000
Voltage c Factor = 1.10 (User-Defined)
Peak Value = 12.573 kA Method C
Steady State = 4.330 kA rms

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus PLN-A | Total | 0.00 | 0.740 | -5.228 | 7.1 | 5.280 |
| 20 kV MVP-L1-CSR-A | Bus PLN-A | 0.04 | 0.310 | -0.919 | 3.0 | 0.970 |
| PLN A | Bus PLN-A | 110.00 | 0.431 | -4.309 | 10.0 | 4.330 |
| Bus3 | 20 kV MVP-L1-CSR-A | 0.05 | 0.050 | -0.126 | 2.5 | 0.135 |
| Bus5 | 20 kV MVP-L1-CSR-A | 0.05 | 0.053 | -0.131 | 2.5 | 0.141 |
| Bus15 | 20 kV MVP-L1-CSR-A | 0.09 | 0.206 | -0.662 | 3.2 | 0.694 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 5.236 | 7.301 | 5.087 |
| 0.02 | 5.213 | 6.383 | 3.682 |
| 0.03 | 5.172 | 5.782 | 2.586 |
| 0.04 | 5.130 | 5.442 | 1.816 |
| 0.05 | 5.077 | 5.257 | 1.366 |
| 0.06 | 5.048 | 5.141 | 0.973 |
| 0.07 | 5.019 | 5.067 | 0.693 |
| 0.08 | 4.991 | 5.016 | 0.493 |
| 0.09 | 4.964 | 4.976 | 0.351 |
| 0.10 | 4.937 | 4.944 | 0.267 |
| 0.15 | 4.878 | 4.878 | 0.050 |
| 0.20 | 4.821 | 4.821 | 0.010 |
| 0.25 | 4.767 | 4.767 | 0.002 |
| 0.30 | 4.767 | 4.767 | 0.000 |

3-Phase fault at bus: **Bus2**

| | | | |
|---|---|---|---|
| Nominal kV | = | 0.400 | |
| Voltage c Factor | = | 1.05 | (User-Defined) |
| Peak Value | = | 78.792 | kA Method C |
| Steady State | = | 30.490 | kA rms |

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus2 | Total | 0.00 | 11.197 | -37.378 | 3.3 | 39.020 |
| Bus4 | Bus2 | 17.44 | 8.145 | -30.113 | 3.7 | 31.195 |
| UDB-L10-A3 | Bus2 | 105.00 | 3.053 | -7.266 | 2.4 | 7.881 |
| Bus3 | Bus4 | 91.76 | 8.145 | -30.113 | 3.7 | 31.195 |

**Breaking and DC Fault Current (kA)**

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 38.363 | 44.388 | 22.330 |
| 0.02 | 38.091 | 39.213 | 9.311 |
| 0.03 | 37.291 | 37.487 | 3.825 |
| 0.04 | 36.522 | 36.555 | 1.571 |
| 0.05 | 35.788 | 35.794 | 0.666 |
| 0.06 | 35.440 | 35.441 | 0.275 |
| 0.07 | 35.100 | 35.101 | 0.114 |
| 0.08 | 34.770 | 34.770 | 0.047 |
| 0.09 | 34.449 | 34.449 | 0.019 |
| 0.10 | 34.138 | 34.138 | 0.008 |
| 0.15 | 33.503 | 33.503 | 0.000 |
| 0.20 | 32.903 | 32.903 | 0.000 |
| 0.25 | 32.340 | 32.340 | 0.000 |
| 0.30 | 32.340 | 32.340 | 0.000 |

3-Phase fault at bus: **Bus3**

Nominal kV = 20.000
Voltage c Factor = 1.10 (User-Defined)
Peak Value = 12.513 kA Method C
Steady State = 4.320 kA rms

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus3 | Total | 0.00 | 0.750 | -5.215 | 7.0 | 5.269 |
| 20 kV MVP-L1-CSR-A | Bus3 | 0.19 | 0.700 | -5.089 | 7.3 | 5.137 |
| Bus4 | Bus3 | 16.80 | 0.050 | -0.126 | 2.5 | 0.135 |
| Bus PLN-A | 20 kV MVP-L1-CSR-A | 0.37 | 0.440 | -4.297 | 9.8 | 4.319 |
| Bus5 | 20 kV MVP-L1-CSR-A | 0.20 | 0.053 | -0.131 | 2.4 | 0.141 |
| Bus15 | 20 kV MVP-L1-CSR-A | 0.23 | 0.207 | -0.662 | 3.2 | 0.693 |
| Bus2 | Bus4 | 19.91 | 2.492 | -6.294 | 2.5 | 6.770 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 5.226 | 7.260 | 5.040 |
| 0.02 | 5.203 | 6.339 | 3.621 |
| 0.03 | 5.161 | 5.745 | 2.524 |
| 0.04 | 5.119 | 5.413 | 1.760 |
| 0.05 | 5.066 | 5.234 | 1.314 |
| 0.06 | 5.037 | 5.122 | 0.928 |
| 0.07 | 5.009 | 5.051 | 0.656 |
| 0.08 | 4.981 | 5.002 | 0.464 |
| 0.09 | 4.953 | 4.964 | 0.328 |
| 0.10 | 4.926 | 4.932 | 0.247 |
| 0.15 | 4.867 | 4.867 | 0.045 |
| 0.20 | 4.810 | 4.810 | 0.008 |
| 0.25 | 4.756 | 4.756 | 0.001 |
| 0.30 | 4.756 | 4.756 | 0.000 |

3-Phase fault at bus: **Bus4**

| | | | |
|---|---|---|---|
| Nominal kV | = | 0.400 | |
| Voltage c Factor | = | 1.05 | (User-Defined) |
| Peak Value | = | 96.725 | kA Method C |
| Steady State | = | 35.119 | kA rms |

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus4 | Total | 0.00 | 8.842 | -42.569 | 4.8 | 43.478 |
| Bus2 | Bus4 | 4.24 | 3.058 | -6.947 | 2.3 | 7.591 |
| Bus3 | Bus4 | 89.55 | 5.784 | -35.622 | 6.2 | 36.088 |
| UDB-L10-A3 | Bus2 | 105.00 | 3.058 | -6.947 | 2.3 | 7.591 |
| 20 kV MVP-L1-CSR-A | Bus3 | 89.57 | 0.116 | -0.712 | 6.2 | 0.722 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 42.895 | 55.211 | 34.760 |
| 0.02 | 42.655 | 47.492 | 20.881 |
| 0.03 | 41.932 | 43.662 | 12.169 |
| 0.04 | 41.236 | 41.841 | 7.092 |
| 0.05 | 40.571 | 40.808 | 4.393 |
| 0.06 | 40.256 | 40.339 | 2.591 |
| 0.07 | 39.948 | 39.978 | 1.529 |
| 0.08 | 39.649 | 39.659 | 0.902 |
| 0.09 | 39.358 | 39.362 | 0.532 |
| 0.10 | 39.075 | 39.077 | 0.331 |
| 0.15 | 38.496 | 38.496 | 0.024 |
| 0.20 | 37.949 | 37.949 | 0.002 |
| 0.25 | 37.436 | 37.436 | 0.000 |
| 0.30 | 37.436 | 37.436 | 0.000 |

3-Phase fault at bus: **Bus5**

Nominal kV = 20.000
Voltage c Factor = 1.10 (User-Defined)
Peak Value = 12.513 kA Method C
Steady State = 4.320 kA rms

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus5 | Total | 0.00 | 0.750 | -5.215 | 7.0 | 5.269 |
| 20 kV MVP-L1-CSR-A | Bus5 | 0.19 | 0.697 | -5.084 | 7.3 | 5.132 |
| Bus6 | Bus5 | 12.53 | 0.053 | -0.131 | 2.5 | 0.141 |
| Bus PLN-A | 20 kV MVP-L1-CSR-A | 0.37 | 0.440 | -4.297 | 9.8 | 4.319 |
| Bus3 | 20 kV MVP-L1-CSR-A | 0.20 | 0.050 | -0.126 | 2.5 | 0.135 |
| Bus15 | 20 kV MVP-L1-CSR-A | 0.23 | 0.207 | -0.662 | 3.2 | 0.693 |
| Bus7 | Bus6 | 15.83 | 2.666 | -6.549 | 2.5 | 7.071 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 5.226 | 7.260 | 5.040 |
| 0.02 | 5.203 | 6.339 | 3.621 |
| 0.03 | 5.161 | 5.745 | 2.524 |
| 0.04 | 5.119 | 5.413 | 1.760 |
| 0.05 | 5.066 | 5.234 | 1.314 |
| 0.06 | 5.037 | 5.122 | 0.928 |
| 0.07 | 5.009 | 5.051 | 0.656 |
| 0.08 | 4.981 | 5.002 | 0.464 |
| 0.09 | 4.953 | 4.964 | 0.328 |
| 0.10 | 4.926 | 4.932 | 0.247 |
| 0.15 | 4.867 | 4.867 | 0.045 |
| 0.20 | 4.810 | 4.810 | 0.008 |
| 0.25 | 4.756 | 4.756 | 0.001 |
| 0.30 | 4.756 | 4.756 | 0.000 |

3-Phase fault at bus: **Bus6**

| | | | |
|---|---|---|---|
| Nominal kV | = | 0.400 | |
| Voltage c Factor | = | 1.05 | (User-Defined) |
| Peak Value | = | 123.654 | kA Method C |
| Steady State | = | 46.035 | kA rms |

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus6 | Total | 0.00 | 10.636 | -54.047 | 5.1 | 55.083 |
| Bus7 | Bus6 | 4.24 | 3.058 | -6.947 | 2.3 | 7.591 |
| Bus5 | Bus6 | 84.56 | 7.578 | -47.100 | 6.2 | 47.705 |
| UDB-L10-A1 | Bus7 | 105.00 | 3.058 | -6.947 | 2.3 | 7.591 |
| 20 kV MVP-L1-CSR-A | Bus5 | 84.58 | 0.152 | -0.942 | 6.2 | 0.954 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 54.503 | 70.707 | 45.043 |
| 0.02 | 54.263 | 60.811 | 27.450 |
| 0.03 | 53.541 | 55.966 | 16.295 |
| 0.04 | 52.847 | 53.725 | 9.673 |
| 0.05 | 52.184 | 52.534 | 6.057 |
| 0.06 | 51.869 | 51.996 | 3.634 |
| 0.07 | 51.562 | 51.608 | 2.180 |
| 0.08 | 51.263 | 51.280 | 1.308 |
| 0.09 | 50.972 | 50.978 | 0.785 |
| 0.10 | 50.690 | 50.692 | 0.494 |
| 0.15 | 50.112 | 50.112 | 0.039 |
| 0.20 | 49.565 | 49.565 | 0.003 |
| 0.25 | 49.052 | 49.052 | 0.000 |
| 0.30 | 49.052 | 49.052 | 0.000 |

3-Phase fault at bus: **Bus7**

| | | | |
|---|---|---|---|
| Nominal kV | = | 0.400 | |
| Voltage c Factor | = | 1.05 | (User-Defined) |
| Peak Value | = | 93.839 | kA Method C |
| Steady State | = | 38.342 | kA rms |

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus7 | Total | 0.00 | 14.347 | -45.062 | 3.1 | 47.291 |
| Bus6 | Bus7 | 22.05 | 11.294 | -37.796 | 3.3 | 39.448 |
| UDB-L10-A1 | Bus7 | 105.00 | 3.053 | -7.266 | 2.4 | 7.881 |
| Bus5 | Bus6 | 88.33 | 11.294 | -37.796 | 3.3 | 39.448 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 46.632 | 53.036 | 25.264 |
| 0.02 | 46.360 | 47.379 | 9.773 |
| 0.03 | 45.559 | 45.712 | 3.736 |
| 0.04 | 44.788 | 44.811 | 1.428 |
| 0.05 | 44.052 | 44.056 | 0.560 |
| 0.06 | 43.704 | 43.704 | 0.215 |
| 0.07 | 43.364 | 43.364 | 0.083 |
| 0.08 | 43.033 | 43.033 | 0.032 |
| 0.09 | 42.711 | 42.711 | 0.012 |
| 0.10 | 42.399 | 42.399 | 0.005 |
| 0.15 | 41.762 | 41.762 | 0.000 |
| 0.20 | 41.161 | 41.161 | 0.000 |
| 0.25 | 40.596 | 40.596 | 0.000 |
| 0.30 | 40.596 | 40.596 | 0.000 |

3-Phase fault at bus: **Bus8**

Nominal kV = 0.400
Voltage c Factor = 1.05 (User-Defined)
Peak Value = 126.194 kA Method C
Steady State = 41.405 kA rms

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus8 | Total | 0.00 | 10.119 | -54.593 | 5.4 | 55.523 |
| LDMP-L2-C1 | Bus8 | 1.28 | 2.170 | -5.096 | 2.3 | 5.539 |
| Bus11 | Bus8 | 59.22 | 7.948 | -49.497 | 6.2 | 50.131 |
| LDMP-L2-CH1 | LDMP-L2-C1 | 105.00 | 2.170 | -5.096 | 2.3 | 5.539 |
| MVSB-L2-CH1 | Bus11 | 59.32 | 0.289 | -1.800 | 6.2 | 1.823 |

**Breaking and DC Fault Current (kA)**

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 54.996 | 72.495 | 47.233 |
| 0.02 | 54.592 | 62.487 | 30.403 |
| 0.03 | 53.971 | 57.191 | 18.919 |
| 0.04 | 53.357 | 54.640 | 11.772 |
| 0.05 | 52.552 | 53.142 | 7.899 |
| 0.06 | 52.103 | 52.341 | 4.990 |
| 0.07 | 51.657 | 51.753 | 3.152 |
| 0.08 | 51.215 | 51.254 | 1.991 |
| 0.09 | 50.779 | 50.794 | 1.258 |
| 0.10 | 50.347 | 50.354 | 0.852 |
| 0.15 | 49.371 | 49.371 | 0.089 |
| 0.20 | 48.415 | 48.415 | 0.009 |
| 0.25 | 47.484 | 47.484 | 0.001 |
| 0.30 | 47.484 | 47.484 | 0.000 |

3-Phase fault at bus: **Bus9**

Nominal kV = 0.400
Voltage c Factor = 1.05 (User-Defined)
Peak Value = 52.543 kA Method C
Steady State = 19.759 kA rms

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus9 | Total | 0.00 | 5.712 | -23.891 | 4.2 | 24.565 |
| Bus10 | Bus9 | 9.79 | 4.552 | -21.131 | 4.6 | 21.615 |
| UDB-L10-A2 | Bus9 | 105.00 | 1.160 | -2.761 | 2.4 | 2.995 |
| Bus20 | Bus10 | 85.66 | 4.552 | -21.131 | 4.6 | 21.615 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 24.220 | 29.713 | 17.211 |
| 0.02 | 23.928 | 25.530 | 8.901 |
| 0.03 | 23.645 | 24.071 | 4.505 |
| 0.04 | 23.375 | 23.485 | 2.280 |
| 0.05 | 23.118 | 23.149 | 1.210 |
| 0.06 | 22.994 | 23.003 | 0.618 |
| 0.07 | 22.875 | 22.877 | 0.316 |
| 0.08 | 22.758 | 22.759 | 0.161 |
| 0.09 | 22.646 | 22.646 | 0.082 |
| 0.10 | 22.536 | 22.536 | 0.044 |
| 0.15 | 22.323 | 22.323 | 0.002 |
| 0.20 | 22.122 | 22.122 | 0.000 |
| 0.25 | 21.935 | 21.935 | 0.000 |
| 0.30 | 21.935 | 21.935 | 0.000 |

3-Phase fault at bus: **Bus10**

| | | | |
|---|---|---|---|
| Nominal kV | = | 0.400 | |
| Voltage c Factor | = | 1.05 | (User-Defined) |
| Peak Value | = | 59.019 | kA Method C |
| Steady State | = | 21.438 | kA rms |

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus10 | Total | 0.00 | 5.277 | -26.018 | 4.9 | 26.547 |
| Bus9 | Bus10 | 1.34 | 1.152 | -2.724 | 2.4 | 2.957 |
| Bus20 | Bus10 | 83.86 | 4.125 | -23.294 | 5.6 | 23.656 |
| UDB-L10-A2 | Bus9 | 105.00 | 1.152 | -2.724 | 2.4 | 2.957 |
| MVSB-L4-A1 | Bus20 | 83.96 | 0.150 | -0.847 | 5.6 | 0.860 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 26.216 | 33.602 | 21.020 |
| 0.02 | 25.933 | 28.726 | 12.356 |
| 0.03 | 25.660 | 26.621 | 7.088 |
| 0.04 | 25.398 | 25.722 | 4.067 |
| 0.05 | 25.150 | 25.270 | 2.459 |
| 0.06 | 25.031 | 25.071 | 1.425 |
| 0.07 | 24.915 | 24.929 | 0.826 |
| 0.08 | 24.802 | 24.807 | 0.479 |
| 0.09 | 24.693 | 24.695 | 0.278 |
| 0.10 | 24.587 | 24.588 | 0.169 |
| 0.15 | 24.381 | 24.381 | 0.011 |
| 0.20 | 24.186 | 24.186 | 0.001 |
| 0.25 | 24.004 | 24.004 | 0.000 |
| 0.30 | 24.004 | 24.004 | 0.000 |

3-Phase fault at bus: **Bus11**

| | | | |
|---|---|---|---|
| Nominal kV | = | 11.000 | |
| Voltage c Factor | = | 1.10 | (User-Defined) |
| Peak Value | = | 9.987 | kA Method C |
| Steady State | = | 2.952 | kA rms |

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus11 | Total | 0.00 | 1.001 | -4.454 | 4.4 | 4.565 |
| MVSB-L2-CH1 | Bus11 | 0.37 | 0.926 | -4.269 | 4.6 | 4.369 |
| Bus8 | Bus11 | 6.47 | 0.075 | -0.184 | 2.5 | 0.199 |
| MVSB-L4-A1 | MVSB-L2-CH1 | 0.98 | 0.926 | -4.269 | 4.6 | 4.369 |
| LDMP-L2-C1 | Bus8 | 7.49 | 2.061 | -5.071 | 2.5 | 5.474 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 4.470 | 5.680 | 3.504 |
| 0.02 | 4.416 | 4.890 | 2.100 |
| 0.03 | 4.348 | 4.510 | 1.198 |
| 0.04 | 4.281 | 4.335 | 0.683 |
| 0.05 | 4.179 | 4.202 | 0.437 |
| 0.06 | 4.121 | 4.129 | 0.255 |
| 0.07 | 4.064 | 4.067 | 0.149 |
| 0.08 | 4.008 | 4.009 | 0.087 |
| 0.09 | 3.953 | 3.954 | 0.051 |
| 0.10 | 3.900 | 3.900 | 0.033 |
| 0.15 | 3.783 | 3.783 | 0.002 |
| 0.20 | 3.671 | 3.671 | 0.000 |
| 0.25 | 3.566 | 3.566 | 0.000 |
| 0.30 | 3.566 | 3.566 | 0.000 |

3-Phase fault at bus: **Bus12**

| | | | |
|---|---|---|---|
| Nominal kV | = | 0.400 | |
| Voltage c Factor | = | 1.05 | (User-Defined) |
| Peak Value | = | 114.482 | kA Method C |
| Steady State | = | 39.261 | kA rms |

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus12 | Total | 0.00 | 9.669 | -50.035 | 5.2 | 50.961 |
| Bus13 | Bus12 | 4.25 | 8.204 | -46.548 | 5.7 | 47.265 |
| UDB-L10-A4 | Bus12 | 105.00 | 1.465 | -3.487 | 2.4 | 3.783 |
| Bus19 | Bus13 | 62.45 | 8.204 | -46.548 | 5.7 | 47.265 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 50.544 | 65.531 | 41.708 |
| 0.02 | 50.124 | 56.267 | 25.563 |
| 0.03 | 49.670 | 51.951 | 15.225 |
| 0.04 | 49.220 | 50.049 | 9.068 |
| 0.05 | 48.601 | 48.942 | 5.763 |
| 0.06 | 48.247 | 48.372 | 3.477 |
| 0.07 | 47.894 | 47.940 | 2.098 |
| 0.08 | 47.545 | 47.561 | 1.266 |
| 0.09 | 47.198 | 47.204 | 0.764 |
| 0.10 | 46.854 | 46.857 | 0.490 |
| 0.15 | 46.072 | 46.072 | 0.040 |
| 0.20 | 45.300 | 45.300 | 0.003 |
| 0.25 | 44.544 | 44.544 | 0.000 |
| 0.30 | 44.544 | 44.544 | 0.000 |

3-Phase fault at bus: **Bus13**

Nominal kV = 0.400
Voltage c Factor = 1.05 (User-Defined)
Peak Value = 120.734 kA Method C
Steady State = 40.490 kA rms

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus13 | Total | 0.00 | 9.175 | -51.948 | 5.7 | 52.752 |
| Bus12 | Bus13 | 0.34 | 1.463 | -3.476 | 2.4 | 3.771 |
| Bus19 | Bus13 | 60.88 | 7.712 | -48.472 | 6.3 | 49.082 |
| UDB-L10-A4 | Bus12 | 105.00 | 1.463 | -3.476 | 2.4 | 3.771 |
| MVSB-L4-A1 | Bus19 | 61.08 | 0.280 | -1.763 | 6.3 | 1.785 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 52.322 | 69.394 | 45.584 |
| 0.02 | 51.897 | 59.714 | 29.538 |
| 0.03 | 51.433 | 54.689 | 18.586 |
| 0.04 | 50.974 | 52.298 | 11.695 |
| 0.05 | 50.331 | 50.941 | 7.862 |
| 0.06 | 49.961 | 50.212 | 5.013 |
| 0.07 | 49.594 | 49.697 | 3.196 |
| 0.08 | 49.229 | 49.271 | 2.038 |
| 0.09 | 48.867 | 48.885 | 1.299 |
| 0.10 | 48.509 | 48.517 | 0.881 |
| 0.15 | 47.691 | 47.691 | 0.096 |
| 0.20 | 46.885 | 46.885 | 0.010 |
| 0.25 | 46.095 | 46.095 | 0.001 |
| 0.30 | 46.095 | 46.095 | 0.000 |

3-Phase fault at bus: **Bus15**

Nominal kV = 20.000
Voltage c Factor = 1.10 (User-Defined)
Peak Value = 12.502 kA Method C
Steady State = 4.315 kA rms

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus15 | Total | 0.00 | 0.749 | -5.211 | 7.0 | 5.264 |
| 20 kV MVP-L1-CSR-A | Bus15 | 0.29 | 0.543 | -4.548 | 8.4 | 4.580 |
| Bus17 | Bus15 | 27.45 | 0.206 | -0.663 | 3.2 | 0.694 |
| Bus PLN-A | 20 kV MVP-L1-CSR-A | 0.48 | 0.440 | -4.292 | 9.8 | 4.314 |
| Bus3 | 20 kV MVP-L1-CSR-A | 0.30 | 0.050 | -0.126 | 2.5 | 0.135 |
| Bus5 | 20 kV MVP-L1-CSR-A | 0.30 | 0.053 | -0.131 | 2.4 | 0.141 |
| MVSB-L4-A1 | Bus17 | 27.52 | 0.366 | -1.175 | 3.2 | 1.231 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 5.221 | 7.253 | 5.035 |
| 0.02 | 5.198 | 6.331 | 3.615 |
| 0.03 | 5.156 | 5.739 | 2.519 |
| 0.04 | 5.115 | 5.407 | 1.755 |
| 0.05 | 5.062 | 5.228 | 1.309 |
| 0.06 | 5.033 | 5.117 | 0.925 |
| 0.07 | 5.004 | 5.046 | 0.653 |
| 0.08 | 4.976 | 4.997 | 0.462 |
| 0.09 | 4.948 | 4.959 | 0.326 |
| 0.10 | 4.921 | 4.928 | 0.245 |
| 0.15 | 4.862 | 4.863 | 0.045 |
| 0.20 | 4.806 | 4.806 | 0.008 |
| 0.25 | 4.751 | 4.751 | 0.001 |
| 0.30 | 4.751 | 4.751 | 0.000 |

3-Phase fault at bus: **Bus16**

| | | | |
|---|---|---|---|
| Nominal kV | = | 0.400 | |
| Voltage c Factor | = | 1.05 | (User-Defined) |
| Peak Value | = | 193.789 | kA Method C |
| Steady State | = | 47.469 | kA rms |

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus16 | Total | 0.00 | 23.208 | -87.766 | 3.8 | 90.782 |
| MV-GSB-01 | Bus16 | 44.85 | 7.944 | -51.438 | 6.5 | 52.048 |
| Loadbank G1 | Bus16 | 105.00 | 15.264 | -36.328 | 2.4 | 39.404 |
| MVSB-L4-A1 | MV-GSB-01 | 44.93 | 0.289 | -1.870 | 6.5 | 1.893 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 87.407 | 109.845 | 66.527 |
| 0.02 | 85.952 | 95.245 | 41.035 |
| 0.03 | 84.344 | 87.476 | 23.199 |
| 0.04 | 82.666 | 83.700 | 13.116 |
| 0.05 | 79.464 | 79.975 | 9.023 |
| 0.06 | 77.524 | 77.705 | 5.306 |
| 0.07 | 75.627 | 75.691 | 3.120 |
| 0.08 | 73.774 | 73.797 | 1.834 |
| 0.09 | 71.968 | 71.976 | 1.079 |
| 0.10 | 70.209 | 70.213 | 0.761 |
| 0.15 | 66.394 | 66.394 | 0.059 |
| 0.20 | 62.780 | 62.780 | 0.005 |
| 0.25 | 59.378 | 59.378 | 0.000 |
| 0.30 | 59.378 | 59.378 | 0.000 |

3-Phase fault at bus: **Bus17**

Nominal kV = 11.000
Voltage c Factor = 1.10 (User-Defined)
Peak Value = 10.127 kA Method C
Steady State = 2.970 kA rms

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus17 | Total | 0.00 | 0.981 | -4.492 | 4.6 | 4.598 |
| MVSB-L4-A1 | Bus17 | 0.10 | 0.535 | -1.486 | 2.8 | 1.579 |
| Bus15 | Bus17 | 66.10 | 0.446 | -3.006 | 6.7 | 3.039 |
| Bus20 | MVSB-L4-A1 | 0.12 | 0.038 | -0.095 | 2.5 | 0.103 |
| Bus19 | MVSB-L4-A1 | 0.12 | 0.052 | -0.127 | 2.5 | 0.138 |
| MVSB-L2-CH1 | MVSB-L4-A1 | 0.13 | 0.075 | -0.184 | 2.5 | 0.199 |
| MV-GSB-01 | MVSB-L4-A1 | 0.19 | 0.370 | -1.079 | 2.9 | 1.141 |
| 20 kV MVP-L1-CSR-A | Bus15 | 66.20 | 0.252 | -1.696 | 6.7 | 1.714 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 4.502 | 5.766 | 3.603 |
| 0.02 | 4.447 | 4.964 | 2.206 |
| 0.03 | 4.379 | 4.564 | 1.285 |
| 0.04 | 4.311 | 4.375 | 0.749 |
| 0.05 | 4.208 | 4.236 | 0.489 |
| 0.06 | 4.149 | 4.159 | 0.292 |
| 0.07 | 4.091 | 4.095 | 0.174 |
| 0.08 | 4.035 | 4.036 | 0.104 |
| 0.09 | 3.980 | 3.980 | 0.062 |
| 0.10 | 3.926 | 3.926 | 0.041 |
| 0.15 | 3.808 | 3.808 | 0.003 |
| 0.20 | 3.695 | 3.695 | 0.000 |
| 0.25 | 3.588 | 3.588 | 0.000 |
| 0.30 | 3.588 | 3.588 | 0.000 |

3-Phase fault at bus: **Bus19**

Nominal kV = 11.000
Voltage c Factor = 1.10 (User-Defined)
Peak Value = 10.029 kA Method C
Steady State = 2.956 kA rms

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus19 | Total | 0.00 | 0.993 | -4.464 | 4.5 | 4.573 |
| MVSB-L4-A1 | Bus19 | 0.65 | 0.941 | -4.337 | 4.6 | 4.438 |
| Bus13 | Bus19 | 4.70 | 0.052 | -0.128 | 2.5 | 0.138 |
| Bus20 | MVSB-L4-A1 | 0.67 | 0.038 | -0.095 | 2.5 | 0.102 |
| MVSB-L2-CH1 | MVSB-L4-A1 | 0.68 | 0.075 | -0.183 | 2.4 | 0.198 |
| Bus17 | MVSB-L4-A1 | 0.85 | 0.456 | -2.985 | 6.5 | 3.020 |
| MV-GSB-01 | MVSB-L4-A1 | 0.75 | 0.372 | -1.074 | 2.9 | 1.136 |
| Bus12 | Bus13 | 5.01 | 1.428 | -3.509 | 2.5 | 3.788 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 4.478 | 5.705 | 3.536 |
| 0.02 | 4.423 | 4.911 | 2.134 |
| 0.03 | 4.356 | 4.525 | 1.226 |
| 0.04 | 4.288 | 4.346 | 0.704 |
| 0.05 | 4.186 | 4.211 | 0.454 |
| 0.06 | 4.128 | 4.137 | 0.267 |
| 0.07 | 4.071 | 4.074 | 0.157 |
| 0.08 | 4.015 | 4.016 | 0.092 |
| 0.09 | 3.960 | 3.960 | 0.054 |
| 0.10 | 3.906 | 3.906 | 0.035 |
| 0.15 | 3.789 | 3.789 | 0.003 |
| 0.20 | 3.677 | 3.677 | 0.000 |
| 0.25 | 3.571 | 3.571 | 0.000 |
| 0.30 | 3.571 | 3.571 | 0.000 |

3-Phase fault at bus: **Bus20**

Nominal kV = 11.000
Voltage c Factor = 1.10 (User-Defined)
Peak Value = 10.027 kA Method C
Steady State = 2.956 kA rms

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| Bus20 | Total | 0.00 | 0.994 | -4.464 | 4.5 | 4.573 |
| MVSB-L4-A1 | Bus20 | 0.66 | 0.956 | -4.368 | 4.6 | 4.472 |
| Bus10 | Bus20 | 10.01 | 0.038 | -0.095 | 2.5 | 0.103 |
| Bus19 | MVSB-L4-A1 | 0.68 | 0.052 | -0.127 | 2.4 | 0.137 |
| MVSB-L2-CH1 | MVSB-L4-A1 | 0.69 | 0.075 | -0.183 | 2.4 | 0.198 |
| Bus17 | MVSB-L4-A1 | 0.86 | 0.456 | -2.985 | 6.5 | 3.019 |
| MV-GSB-01 | MVSB-L4-A1 | 0.76 | 0.372 | -1.074 | 2.9 | 1.136 |
| Bus9 | Bus10 | 11.19 | 1.044 | -2.623 | 2.5 | 2.823 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 4.477 | 5.705 | 3.535 |
| 0.02 | 4.423 | 4.911 | 2.133 |
| 0.03 | 4.356 | 4.525 | 1.225 |
| 0.04 | 4.288 | 4.345 | 0.704 |
| 0.05 | 4.186 | 4.211 | 0.453 |
| 0.06 | 4.128 | 4.136 | 0.266 |
| 0.07 | 4.071 | 4.074 | 0.157 |
| 0.08 | 4.015 | 4.016 | 0.092 |
| 0.09 | 3.960 | 3.960 | 0.054 |
| 0.10 | 3.906 | 3.906 | 0.035 |
| 0.15 | 3.789 | 3.789 | 0.003 |
| 0.20 | 3.677 | 3.677 | 0.000 |
| 0.25 | 3.571 | 3.571 | 0.000 |
| 0.30 | 3.571 | 3.571 | 0.000 |

3-Phase fault at bus: **LDMP-L2-C1**

| | | | |
|---|---|---|---|
| Nominal kV | = | 0.400 | |
| Voltage c Factor | = | 1.05 | (User-Defined) |
| Peak Value | = | 109.853 | kA Method C |
| Steady State | = | 38.601 | kA rms |

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| LDMP-L2-C1 | Total | 0.00 | 12.228 | -50.042 | 4.1 | 51.514 |
| Bus8 | LDMP-L2-C1 | 10.64 | 10.058 | -44.878 | 4.5 | 45.991 |
| LDMP-L2-CH1 | LDMP-L2-C1 | 105.00 | 2.170 | -5.164 | 2.4 | 5.601 |
| Bus11 | Bus8 | 62.89 | 10.058 | -44.878 | 4.5 | 45.991 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 51.010 | 62.405 | 35.951 |
| 0.02 | 50.618 | 54.010 | 18.840 |
| 0.03 | 50.004 | 50.914 | 9.581 |
| 0.04 | 49.400 | 49.639 | 4.872 |
| 0.05 | 48.637 | 48.709 | 2.654 |
| 0.06 | 48.217 | 48.236 | 1.368 |
| 0.07 | 47.800 | 47.805 | 0.705 |
| 0.08 | 47.388 | 47.389 | 0.364 |
| 0.09 | 46.980 | 46.980 | 0.187 |
| 0.10 | 46.577 | 46.577 | 0.103 |
| 0.15 | 45.672 | 45.672 | 0.004 |
| 0.20 | 44.786 | 44.786 | 0.000 |
| 0.25 | 43.923 | 43.923 | 0.000 |
| 0.30 | 43.923 | 43.923 | 0.000 |

3-Phase fault at bus: **MV-GSB-01**

Nominal kV = 11.000
Voltage c Factor = 1.10 (User-Defined)
Peak Value = 10.085 kA Method C
Steady State = 2.962 kA rms

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| MV-GSB-01 | Total | 0.00 | 0.989 | -4.483 | 4.5 | 4.591 |
| MVSB-L4-A1 | MV-GSB-01 | 0.28 | 0.620 | -3.402 | 5.5 | 3.458 |
| Bus16 | MV-GSB-01 | 27.08 | 0.370 | -1.081 | 2.9 | 1.143 |
| Bus20 | MVSB-L4-A1 | 0.30 | 0.038 | -0.095 | 2.5 | 0.102 |
| Bus19 | MVSB-L4-A1 | 0.30 | 0.052 | -0.127 | 2.4 | 0.138 |
| MVSB-L2-CH1 | MVSB-L4-A1 | 0.31 | 0.075 | -0.184 | 2.4 | 0.199 |
| Bus17 | MVSB-L4-A1 | 0.47 | 0.454 | -2.996 | 6.6 | 3.030 |
| Loadbank G1 | Bus16 | 110.00 | 10.168 | -29.730 | 2.9 | 31.421 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 4.495 | 5.737 | 3.566 |
| 0.02 | 4.440 | 4.937 | 2.159 |
| 0.03 | 4.372 | 4.546 | 1.245 |
| 0.04 | 4.303 | 4.363 | 0.718 |
| 0.05 | 4.201 | 4.226 | 0.464 |
| 0.06 | 4.142 | 4.151 | 0.273 |
| 0.07 | 4.084 | 4.087 | 0.161 |
| 0.08 | 4.027 | 4.028 | 0.095 |
| 0.09 | 3.972 | 3.972 | 0.056 |
| 0.10 | 3.918 | 3.918 | 0.037 |
| 0.15 | 3.800 | 3.800 | 0.003 |
| 0.20 | 3.687 | 3.687 | 0.000 |
| 0.25 | 3.580 | 3.580 | 0.000 |
| 0.30 | 3.580 | 3.580 | 0.000 |

3-Phase fault at bus: **MVSB-L2-CH1**

| | | | |
|---|---|---|---|
| Nominal kV | = | 11.000 | |
| Voltage c Factor | = | 1.10 | (User-Defined) |
| Peak Value | = | 10.035 | kA Method C |
| Steady State | = | 2.957 | kA rms |

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| MVSB-L2-CH1 | Total | 0.00 | 0.993 | -4.466 | 4.5 | 4.575 |
| MVSB-L4-A1 | MVSB-L2-CH1 | 0.61 | 0.918 | -4.282 | 4.7 | 4.379 |
| Bus11 | MVSB-L2-CH1 | 0.02 | 0.075 | -0.184 | 2.5 | 0.199 |
| Bus20 | MVSB-L4-A1 | 0.63 | 0.038 | -0.095 | 2.5 | 0.102 |
| Bus19 | MVSB-L4-A1 | 0.63 | 0.052 | -0.127 | 2.4 | 0.137 |
| Bus17 | MVSB-L4-A1 | 0.81 | 0.456 | -2.986 | 6.6 | 3.021 |
| MV-GSB-01 | MVSB-L4-A1 | 0.70 | 0.372 | -1.074 | 2.9 | 1.137 |
| Bus8 | Bus11 | 6.48 | 0.075 | -0.184 | 2.5 | 0.199 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 4.480 | 5.709 | 3.539 |
| 0.02 | 4.425 | 4.914 | 2.138 |
| 0.03 | 4.358 | 4.528 | 1.229 |
| 0.04 | 4.290 | 4.347 | 0.706 |
| 0.05 | 4.188 | 4.212 | 0.455 |
| 0.06 | 4.129 | 4.138 | 0.268 |
| 0.07 | 4.072 | 4.075 | 0.157 |
| 0.08 | 4.016 | 4.017 | 0.093 |
| 0.09 | 3.961 | 3.961 | 0.054 |
| 0.10 | 3.907 | 3.907 | 0.036 |
| 0.15 | 3.790 | 3.790 | 0.003 |
| 0.20 | 3.678 | 3.678 | 0.000 |
| 0.25 | 3.571 | 3.571 | 0.000 |
| 0.30 | 3.571 | 3.571 | 0.000 |

3-Phase fault at bus: **MVSB-L4-A1**

| | | | |
|---|---|---|---|
| Nominal kV | = | 11.000 | |
| Voltage c Factor | = | 1.10 | (User-Defined) |
| Peak Value | = | 10.112 | kA Method C |
| Steady State | = | 2.966 | kA rms |

| Contribution | | Voltage & Initial Symmetrical Current (rms) | | | | |
|---|---|---|---|---|---|---|
| From Bus ID | To Bus ID | % V From Bus | kA Real | kA Imaginary | X/R Ratio | kA Magnitude |
| MVSB-L4-A1 | Total | 0.00 | 0.984 | -4.489 | 4.6 | 4.595 |
| Bus20 | MVSB-L4-A1 | 0.02 | 0.038 | -0.095 | 2.5 | 0.103 |
| Bus19 | MVSB-L4-A1 | 0.02 | 0.052 | -0.128 | 2.5 | 0.138 |
| MVSB-L2-CH1 | MVSB-L4-A1 | 0.03 | 0.075 | -0.184 | 2.5 | 0.199 |
| Bus17 | MVSB-L4-A1 | 0.20 | 0.448 | -3.001 | 6.7 | 3.035 |
| MV-GSB-01 | MVSB-L4-A1 | 0.09 | 0.370 | -1.080 | 2.9 | 1.142 |
| Bus10 | Bus20 | 10.02 | 0.038 | -0.095 | 2.5 | 0.103 |
| Bus13 | Bus19 | 4.71 | 0.052 | -0.128 | 2.5 | 0.138 |
| Bus11 | MVSB-L2-CH1 | 0.04 | 0.075 | -0.184 | 2.5 | 0.199 |
| Bus15 | Bus17 | 66.17 | 0.448 | -3.001 | 6.7 | 3.035 |
| Bus16 | MV-GSB-01 | 27.11 | 0.370 | -1.080 | 2.9 | 1.142 |

## Breaking and DC Fault Current (kA)

**Based on Total Bus Fault Current**

| TD (S) | Ib sym | Ib asym | Idc |
|---|---|---|---|
| 0.01 | 4.499 | 5.756 | 3.591 |
| 0.02 | 4.444 | 4.954 | 2.190 |
| 0.03 | 4.376 | 4.557 | 1.272 |
| 0.04 | 4.307 | 4.370 | 0.738 |
| 0.05 | 4.204 | 4.232 | 0.481 |
| 0.06 | 4.145 | 4.155 | 0.285 |
| 0.07 | 4.088 | 4.091 | 0.170 |
| 0.08 | 4.031 | 4.032 | 0.101 |
| 0.09 | 3.976 | 3.976 | 0.060 |
| 0.10 | 3.921 | 3.922 | 0.040 |
| 0.15 | 3.803 | 3.803 | 0.003 |
| 0.20 | 3.690 | 3.690 | 0.000 |
| 0.25 | 3.583 | 3.583 | 0.000 |
| 0.30 | 3.583 | 3.583 | 0.000 |

## Short-Circuit Summary Report

3-Phase Fault Currents

| Bus | | Device | | Device Capacity (kA) | | | | Short-Circuit Current (kA) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Making | | | | | | | | | |
| ID | kV | ID | Type | Peak | Ib sym | Ib asym | Idc | I"k | ip | Ib sym | Ib asym | Idc | Ik |
| 20 kV MVP-L1-CSR-A | 20.000 | 20 kV MVP-L1-CSR-A | Bus | | | | | 5.274 | 12.546 | | | | 4.324 |
| | 20.000 | CB2 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 5.274 | 12.546 | 5.231 | 7.283 | 5.067 | |
| | 20.000 | CB3 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 5.274 | 12.546 | 5.231 | 7.283 | 5.067 | |
| | 20.000 | CB6 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 5.274 | 12.546 | 5.231 | 7.283 | 5.067 | |
| | 20.000 | CB15 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 5.274 | 12.546 | 5.231 | 7.283 | 5.067 | |
| Bus PLN-A | 20.000 | Bus PLN-A | Bus | | | | | 5.280 | 12.573 | | | | 4.330 |
| | 20.000 | CB1 | CB | 63.000 | 25.000 | 37.769 | 28.310 | 5.280 | 12.573 | 5.236 | 7.301 | 5.087 | |
| Bus2 | 0.400 | Bus2 | Bus | | | | | 39.020 | 78.792 | | | | 30.490 |
| | 0.400 | CB5 | CB | 143.000 | 65.000 | 67.581 | | 39.020 | 78.792 | 37.688 | 38.157 | 5.967 | |
| | 0.400 | CB4 | CB | 143.000 | 65.000 | 67.581 | | 39.020 | 78.792 | 37.688 | 38.157 | 5.967 | |
| Bus3 | 20.000 | Bus3 | Bus | | | | | 5.269 | 12.513 | | | | 4.320 |
| Bus4 | 0.400 | Bus4 | Bus | | | | | 43.478 | 96.725 | | | | 35.119 |
| Bus5 | 20.000 | Bus5 | Bus | | | | | 5.269 | 12.513 | | | | 4.320 |
| Bus6 | 0.400 | Bus6 | Bus | | | | | 55.083 | 123.654 | | | | 46.035 |
| Bus7 | 0.400 | Bus7 | Bus | | | | | 47.291 | 93.839 | | | | 38.342 |
| | 0.400 | CB8 | CB | 143.000 | 65.000 | 67.581 | | 47.291 | 93.839 | 45.956 | 46.352 | 6.042 | |
| | 0.400 | CB7 | CB | 143.000 | 65.000 | 67.581 | | 47.291 | 93.839 | 45.956 | 46.352 | 6.042 | |
| Bus8 | 0.400 | Bus8 | Bus | | | | | 55.523 | 126.194 | | | | 41.405 |
| Bus9 | 0.400 | Bus9 | Bus | | | | | 24.565 | 52.543 | | | | 19.759 |
| | 0.400 | CB10 | CB | 94.500 | 45.000 | 45.014 | | 24.565 | 52.543 | 23.118 | 23.149 | 1.210 | |
| | 0.400 | CB11 | CB | 94.500 | 45.000 | 45.014 | | 24.565 | 52.543 | 23.118 | 23.149 | 1.210 | |
| Bus10 | 0.400 | Bus10 | Bus | | | | | 26.547 | 59.019 | | | | 21.438 |
| Bus11 | 11.000 | Bus11 | Bus | | | | | 4.565 | 9.987 | | | | 2.952 |
| Bus12 | 0.400 | Bus12 | Bus | | | | | 50.961 | 114.482 | | | | 39.261 |
| | 0.400 | CB13 | CB | 220.000 | 85.000 | 105.983 | | 50.961 | 114.482 | 50.544 | 65.531 | 41.708 | |
| | 0.400 | CB14 | CB | 220.000 | 85.000 | 105.983 | | 50.961 | 114.482 | 50.544 | 65.531 | 41.708 | |
| Bus13 | 0.400 | Bus13 | Bus | | | | | 52.752 | 120.734 | | | | 40.490 |
| Bus15 | 20.000 | Bus15 | Bus | | | | | 5.264 | 12.502 | | | | 4.315 |
| Bus16 | 0.400 | Bus16 | Bus | | | | | 90.782 | 193.789 | | | | 47.469 |
| Bus17 | 11.000 | Bus17 | Bus | | | | | 4.598 | 10.127 | | | | 2.970 |
| Bus19 | 11.000 | Bus19 | Bus | | | | | 4.573 | 10.029 | | | | 2.956 |
| Bus20 | 11.000 | Bus20 | Bus | | | | | 4.573 | 10.027 | | | | 2.956 |
| LDMP-L2-C1 | 0.400 | LDMP-L2-C1 | Bus | | | | | 51.514 | 109.853 | | | | 38.601 |
| | 0.400 | CB21 | CB | 121.000 | 55.000 | 57.184 | | 51.514 | 109.853 | 50.310 | 52.073 | 13.435 | |
| | 0.400 | CB20 | CB | 121.000 | 55.000 | 57.184 | | 51.514 | 109.853 | 50.310 | 52.073 | 13.435 | |

3-Phase Fault Currents

| Bus | | Device | | Device Capacity (kA) | | | | Short-Circuit Current (kA) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Making | | | | | | | | | |
| ID | kV | ID | Type | Peak | Ib sym | Ib asym | Idc | I"k | ip | Ib sym | Ib asym | Idc | Ik |
| MV-GSB-01 | 11.000 | MV-GSB-01 | Bus | | | | | 4.591 | 10.085 | | | | 2.962 |
| | 11.000 | CB23 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 4.591 | 10.085 | 4.495 | 5.737 | 3.566 | |
| | 11.000 | CB25 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 4.591 | 10.085 | 4.495 | 5.737 | 3.566 | |
| MVSB-L2-CH1 | 11.000 | MVSB-L2-CH1 | Bus | | | | | 4.575 | 10.035 | | | | 2.957 |
| | 11.000 | CB18 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 4.575 | 10.035 | 4.480 | 5.709 | 3.539 | |
| | 11.000 | CB19 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 4.575 | 10.035 | 4.480 | 5.709 | 3.539 | |
| MVSB-L4-A1 | 11.000 | MVSB-L4-A1 | Bus | | | | | 4.595 | 10.112 | | | | 2.966 |
| | 11.000 | CB12 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 4.595 | 10.112 | 4.499 | 5.756 | 3.591 | |
| | 11.000 | CB9 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 4.595 | 10.112 | 4.499 | 5.756 | 3.591 | |
| | 11.000 | CB17 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 4.595 | 10.112 | 4.499 | 5.756 | 3.591 | |
| | 11.000 | CB16 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 4.595 | 10.112 | 4.499 | 5.756 | 3.591 | |
| | 11.000 | CB22 | CB | 40.000 | 16.000 | 24.172 | 18.119 | 4.595 | 10.112 | 4.499 | 5.756 | 3.591 | |

ip is calculated using method C
Ib does not include decay of non-terminal faulted induction motors
Ik is the maximum steady state fault current
Idc is based on X/R from Method C and Ib as specified above

LV CB duty determined based on service rating.
Total through current is used for device duty.

* Indicates a device with calculated duty exceeding the device capability.

\# Indicates a device with calculated duty exceeding the device marginal limit. ( 95 % times device capability)

## Short-Circuit Summary Report

| | | Device Capacity | | | 3-Phase Short-Circuit Duty Results | | |
|---|---|---|---|---|---|---|---|
| Bus ID | Device ID | Ithr (kA) | Tkr (sec.) | Rated Thermal Energy (MJ) | Ith (kA) | Tkr (sec.) | Thermal Energy (MJ) |
| 20 kV MVP-L1-CSR-A | CB2 | 16.000 | 3.00 | 768.00 | 4.713 | 3.00 | 66.64 |
| 20 kV MVP-L1-CSR-A | CB3 | 16.000 | 3.00 | 768.00 | 4.713 | 3.00 | 66.64 |
| 20 kV MVP-L1-CSR-A | CB6 | 16.000 | 3.00 | 768.00 | 4.713 | 3.00 | 66.64 |
| 20 kV MVP-L1-CSR-A | CB15 | 16.000 | 3.00 | 768.00 | 4.713 | 3.00 | 66.64 |
| Bus PLN-A | CB1 | 25.000 | 3.00 | 1875.00 | 4.719 | 3.00 | 66.81 |
| Bus2 | CB5 | 65.000 | 1.00 | 4225.00 | 35.884 | 1.00 | 1287.68 |
| Bus2 | CB4 | 65.000 | 1.00 | 4225.00 | 35.884 | 1.00 | 1287.68 |
| Bus7 | CB8 | 65.000 | 1.00 | 4225.00 | 43.921 | 1.00 | 1929.05 |
| Bus7 | CB7 | 65.000 | 1.00 | 4225.00 | 43.921 | 1.00 | 1929.05 |
| Bus9 | CB10 | 30.000 | 1.00 | 900.00 | 22.817 | 1.00 | 520.61 |
| Bus9 | CB11 | 30.000 | 1.00 | 900.00 | 22.817 | 1.00 | 520.61 |
| Bus12 | CB13 | 45.000 | 3.00 | 6075.00 | 44.104 | 3.00 | 5835.42 |
| Bus12 | CB14 | 45.000 | 3.00 | 6075.00 | 44.104 | 3.00 | 5835.42 |
| LDMP-L2-C1 | CB21 | 55.000 | 1.00 | 3025.00 | 46.892 | 1.00 | 2198.90 |
| LDMP-L2-C1 | CB20 | 55.000 | 1.00 | 3025.00 | 46.892 | 1.00 | 2198.90 |
| MV-GSB-01 | CB23 | 16.000 | 3.00 | 768.00 | 3.622 | 3.00 | 39.35 |
| MV-GSB-01 | CB25 | 16.000 | 3.00 | 768.00 | 3.622 | 3.00 | 39.35 |
| MVSB-L2-CH1 | CB18 | 16.000 | 3.00 | 768.00 | 3.612 | 3.00 | 39.14 |
| MVSB-L2-CH1 | CB19 | 16.000 | 3.00 | 768.00 | 3.612 | 3.00 | 39.14 |
| MVSB-L4-A1 | CB12 | 16.000 | 3.00 | 768.00 | 3.626 | 3.00 | 39.43 |
| MVSB-L4-A1 | CB9 | 16.000 | 3.00 | 768.00 | 3.626 | 3.00 | 39.43 |
| MVSB-L4-A1 | CB17 | 16.000 | 3.00 | 768.00 | 3.626 | 3.00 | 39.43 |
| MVSB-L4-A1 | CB16 | 16.000 | 3.00 | 768.00 | 3.626 | 3.00 | 39.43 |
| MVSB-L4-A1 | CB22 | 16.000 | 3.00 | 768.00 | 3.626 | 3.00 | 39.43 |

Ithr = Rated short-time withstand current (Icw for low voltage circuit breaker)
Tkr = Rated short-time
Ith = Thermal equivalent short-time current

* Indicates a device with calculated duty exceeding the device capability.

\# Indicates a device with calculated duty exceeding the device marginal limit. ( 95 % times device capability)